Term-1

25/11/23 M/S

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[23BSMA12]

FIRST SEMESTER B.Tech EXAMINATION, NOVEMBER - 2023

COMPUTATIONAL MATHEMATICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.
ii. Answer All Questions.

		PART - A			**5 X 2=10**
1.	**a.**	If $A = \{3,4\}$ then, find the power set of A and write its cardinal number.	L-1	CO1	**02**
	b.	Write negation of the following statement. **i.** $p = 2 + 3 > 1$. **ii.** It will rain tomorrow, or it will snow tomorrow.	L-2	CO1	**02**
	c.	Find the first order partial derivatives of $u = e^x \cos y$.	L-2	CO1	**02**
	d.	Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 3 & 5 \end{bmatrix}$.	L-1	CO1	**02**
	e.	Solve the differential equation $(D^2 + 1)y = 0$.	L-2	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Explain different type of relations with example.	L-2	CO1	**05**
		OR			
	b.	Consider $f(x) = x^3 + 1$ and $g(x) = \sqrt[3]{x - 1}$ defined on on f;g: R→ R. Show that $(g\ o\ f)(1) = 1, (g\ o\ f)(2) = 2, (g\ o\ f)(3) = 3, and$ $(g\ o\ f)(x) = x$.	L-3	CO1	**05**
3.	**a.**	Write the truth table for conditional, inverse, converse and contrapositive for two propositions $p\ and\ q$.	L-3	CO3	**05**
		OR			
	b.	Prove the following logical equivalence without using Truth table. $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \Leftrightarrow (p \vee q \vee r)$.	L-3	CO3	**05**
4.	**a.**	If $z = e^{ax+by} f(ax - by)$ then show that $b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz$.	L-3	CO6	**05**
		OR			
	b.	Find the directional derivative of $f = x^2yz + 4xz^2$ at (1, -2, -1) along $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$.	L-3	CO6	**05**

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5.	**a.**	Find the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$ by converting it into Echelon form.	L-3	CO2	**05**
		OR			
	b.	Test the consistency of system of equation $x + y + z = 6$ $x - y + 2z = 5$ $3x + y + z = 8.$	L-3	CO2	**05**
6.	**a.**	Find the General solution of the given differential equation $4y'' + 4y' - 3y = 0.$	L-3	CO5	**05**
		OR			
	b.	Find particular integral (y_p) for the given differential equation $(D^2 + 4)\, y = 4\sin 2x + \cos 5x$	L-3	CO5	**05**
7.	**a.**	Examine whether the function $f(x, y) = x^3 + y^3 - 3xy + 1$ is minimum at the point (1,1).	L-3	CO6	**05**
		OR			
	b.	Find divergence and curl of $\vec{v} = (xyz)i + (3x^2y)j + (xz^2 - y^2z)k$ at (2, -1, 1).	L-3	CO6	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Use mathematical induction to prove that $$\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$ for all integers $n \geq 1$.	L-3	CO1	**10**
		OR			
	b.	Explain various methods by which sets can be created in Python with examples, then write the Python code for set union, intersection, and difference of $S = \{1,2,3,4,5\}$ $and\, T = \{5,6,7,8, 9, 10\}$.	L-4	CO2	**10**

9.	**a.**	Test the validity of the following arguments If Ravi goes out with friends, he will not study. If Ravi does not study, his father becomes angry His father is not angry ∴ Ravi has not gone out with friends	L-3	CO3	**10**
		OR			
	b.	Write a python code for negation, conjunction, disjunction and conditional of two propositions p and q.	L-4	CO3	**10**
10.	**a.**	Prove that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3}{(x+y+z)}$, where $u=log(x^3+y^3+z^3-3xyz)$ and hence show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=\frac{-9}{(x+y+z)^2}$.	L-3	CO6	**10**
		OR			
	b.	Which Python package is used to calculate partial derivatives? Also find all first, second, and third-order partial derivatives of the following function, $f(x,y)=x^4-3x^2y^3$.	L-4	CO6	**10**
11.	**a.**	Find all the eigenvalues and the corresponding eigenvectors of the matrix $\begin{bmatrix}-1 & 3\\ -2 & 4\end{bmatrix}$ hence find A^4.	L-3	CO4	**10**
		OR			
	b.	Write a python code to find all the eigen values and corresponding eigen vectors of $3X3$ matrix. Also write the condition so that two matrices are similar?	L-4	CO2	**10**
12.	**a.**	Solve $X'=\begin{bmatrix}-2 & -1 & 8\\ 0 & -3 & 8\\ 0 & -4 & 9\end{bmatrix}X$ by diagonalization.	L-3	CO5	**10**
		OR			
	b.	Write the python code for solving the differential equation $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+6y=0$, $y(0)=1, y'(0)=-1$.	L-4	CO4	**10**
13.	**a.**	Find the extreme value of the function $f(x,y)=x^3+3xy^2-3x^2-3y^2+4$.	L-3	CO6	**10**
		OR			
	b.	Solve the differential equation $(D^2-D)y=x^2-2x-32$.	L-3	CO5	**10**